DASC 3240, L04

Review of Python basics

2023/24 SPRING SEMESTER

MASAYUKI USHIO, ASSISTANT PROFESSOR AT OCES
USHIO@UST.HK (RM CYT-2013)

Today's outline: Review of Python basics

1. Python
2. Jupyter notebook
3. Git and Github
4. Google Colab
5. How this course uses these platforms
6. Information for assignments
7. Review of Python basics using codes

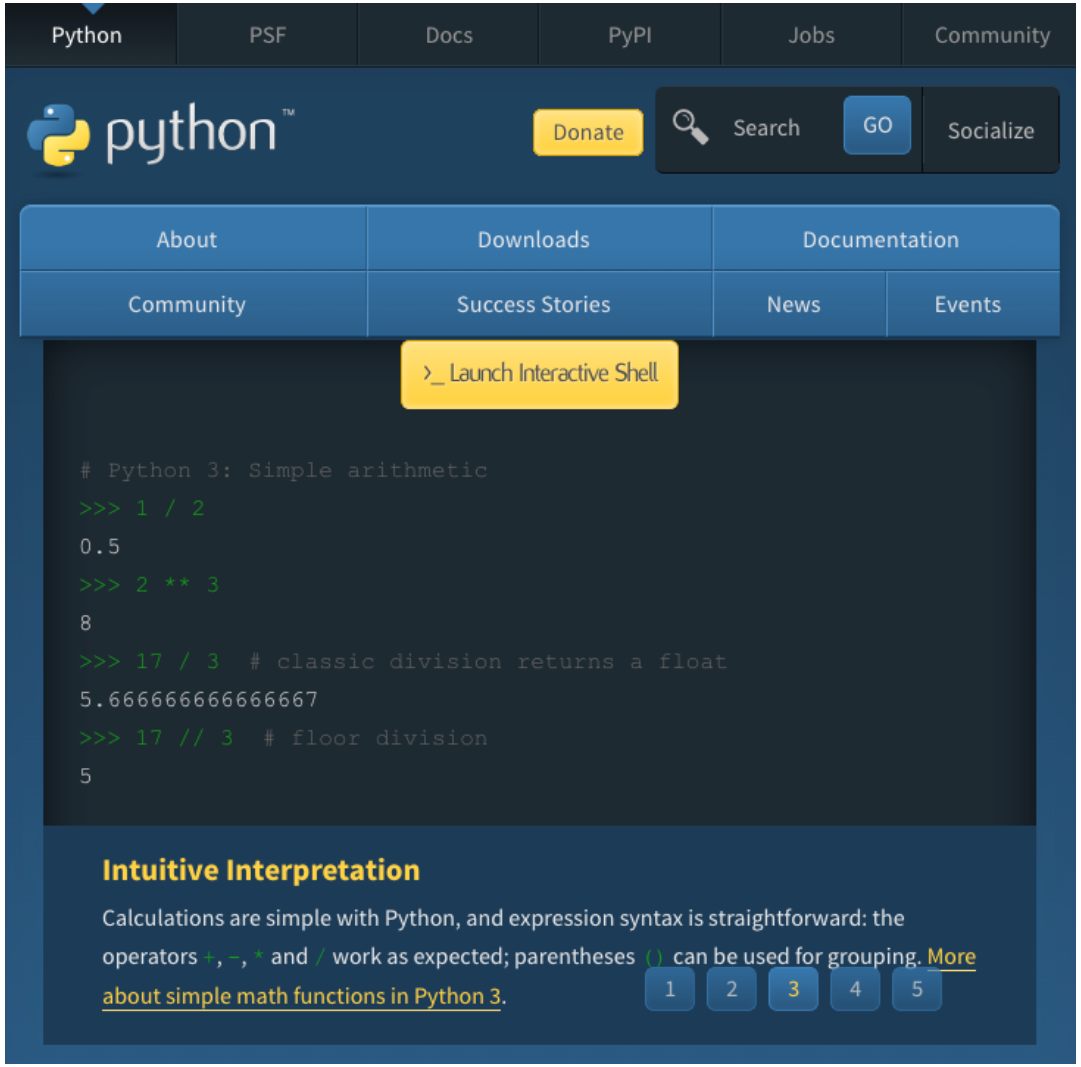
Python

The Basics of Python

Python is a general purpose programming language that supports rapid development of scripts and applications.

Main adbantages:

- Open source, Free
- Multi-platform (Windows, Mac, Linux)
- Very large community (= easy to get help from internet/others)



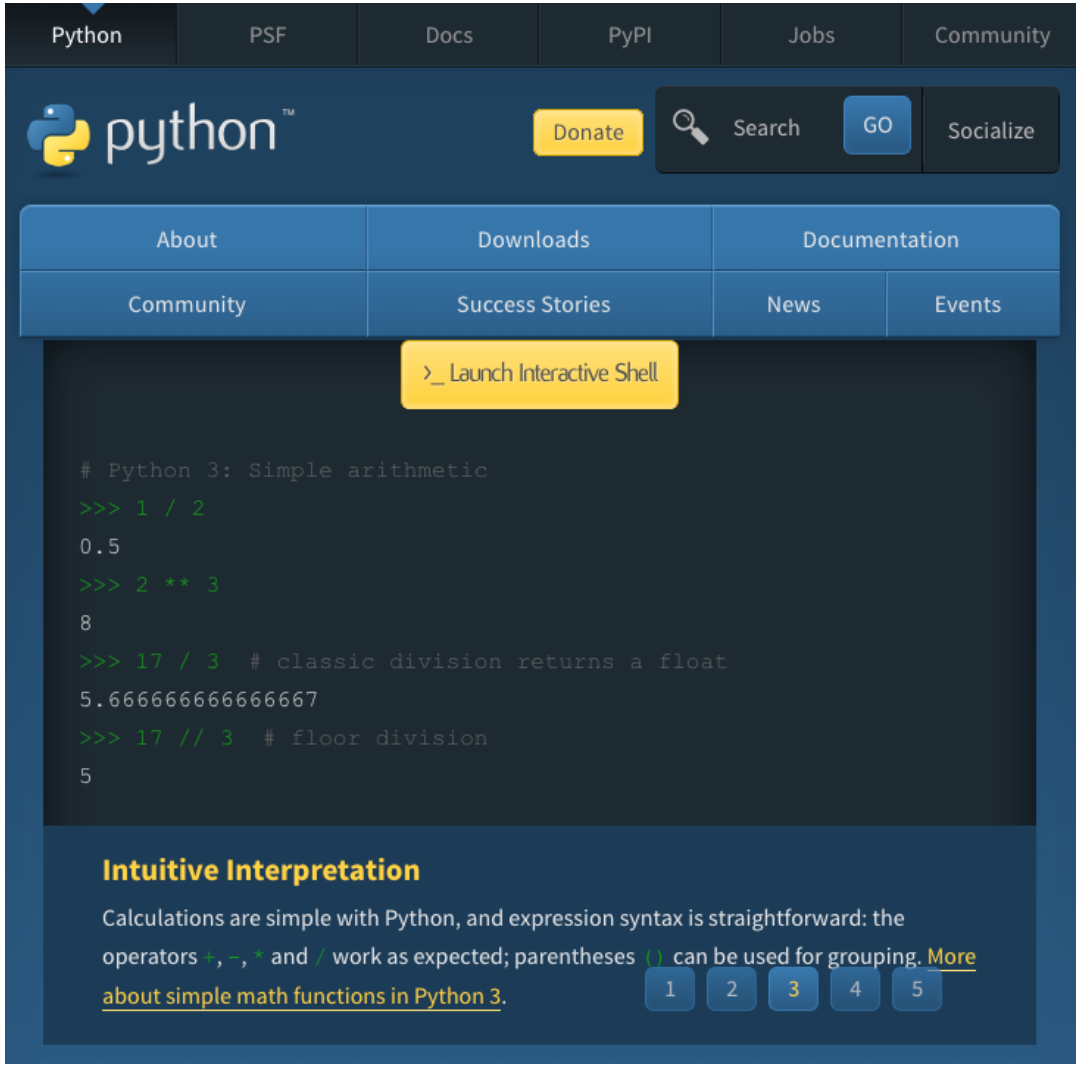
The Basics of Python

Python is an "interpreted language."

What does this mean?

A compiled language = Whole script should be first compiled into machine code. Otherwise the program cannot be executed. **Fast, but probably not for beginners.**

An interpreted language = Executed (translated) line by line. **Slow, but easy to understand and debug.**



Introduction of other tools

Jupyter notebook

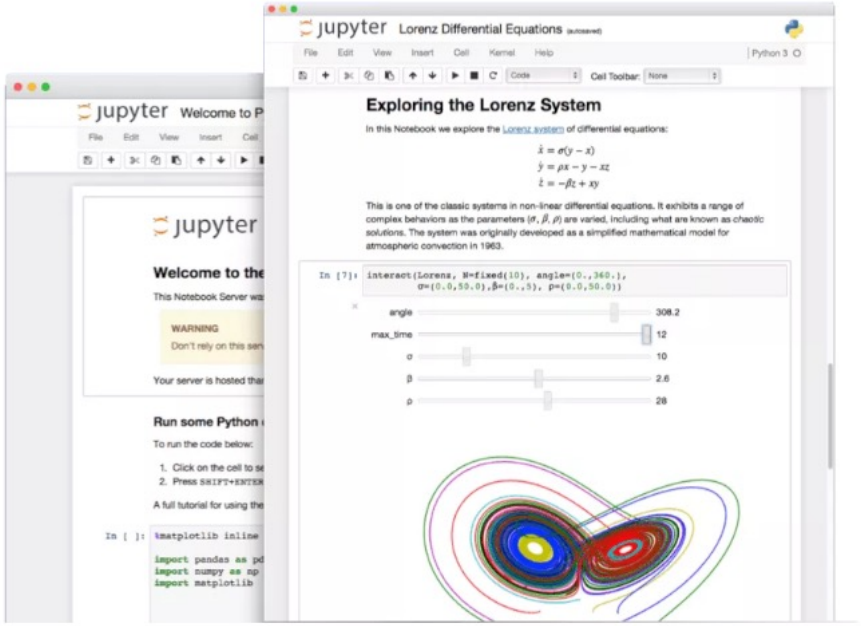
Jupyter

(= Julia + python + R)

Free software, open standards, and web services for interactive computing across all programming languages.

https://jupyter.org/
2015~

In short, you can effectively share your Python code using jupyter notebook.



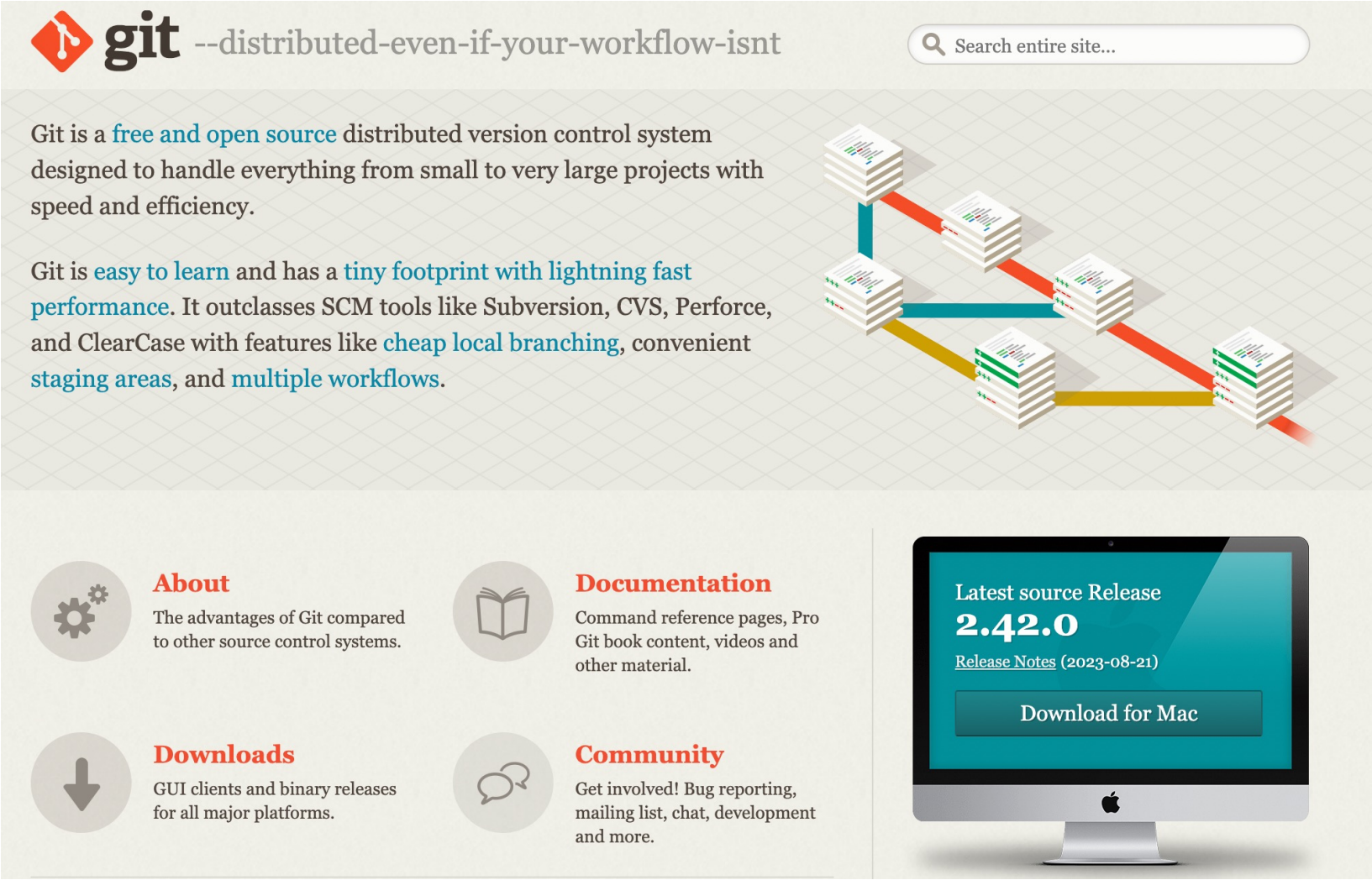
Jupyter Notebook: The Classic Notebook Interface

The Jupyter Notebook is the original web application for creating and sharing computational documents. It offers a simple, streamlined, document-centric experience.

Try it in your browser | Install the Notebook

Git and Github: What is git?

- Version control software
  - Record and track changes in your analysis codes and data.
  - We can check history of the codes; who made changes and when.
  - Revert the histroy if necessary.



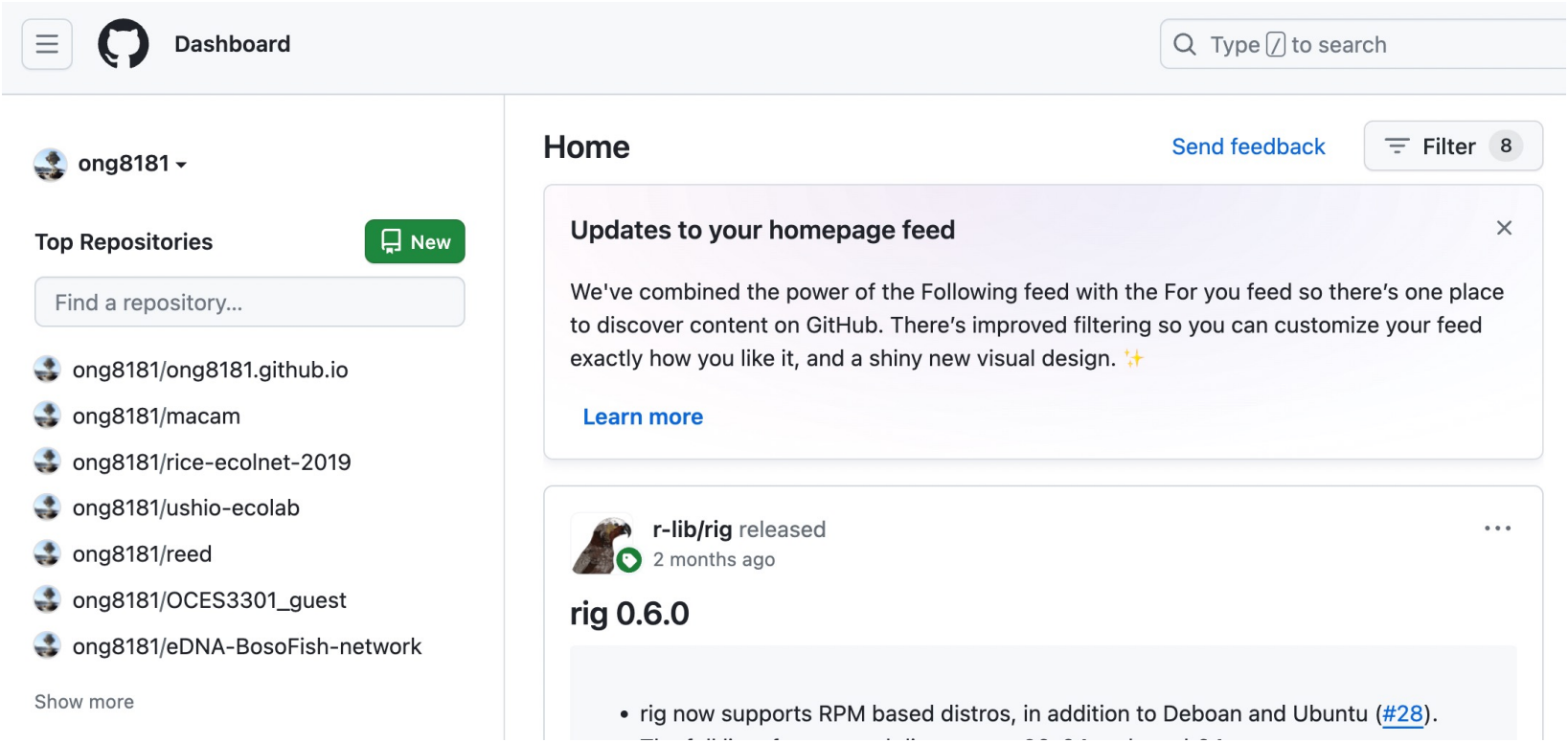
https://git-scm.com/

“What is Git? Explained in 2 Minites”

https://youtu.be/2ReR1YJrNOM?si=xvsma6EpjzYo5k5f

Github: Online platform to share and develop codes

Github allows efficient sharing and development of codes. It has “git” functionality.



“Intro to Google Colab”

https://youtu.be/inN8seMm7UI?si=6FHcpTQk3e-BrtaU

How this course uses these platforms

me | you

Presentation slides

Python codes
Other text resources

upload

Prepared with
jupyter notebook

upload

download is
not necessary

download

GitHub

https://github.com/ong8181/DASC3240

Option 1:
If you do not have Python and jupyter in your PC, you can execute the codes using Google Colab.

colab

Option 2:
If you have Python and jupyter in your PC, you can download the codes and execute them.

If you have a trouble(s), you can utilize a face-to-face assistant opportunity in the tutorial.

※ ChatGPT and other AI tools

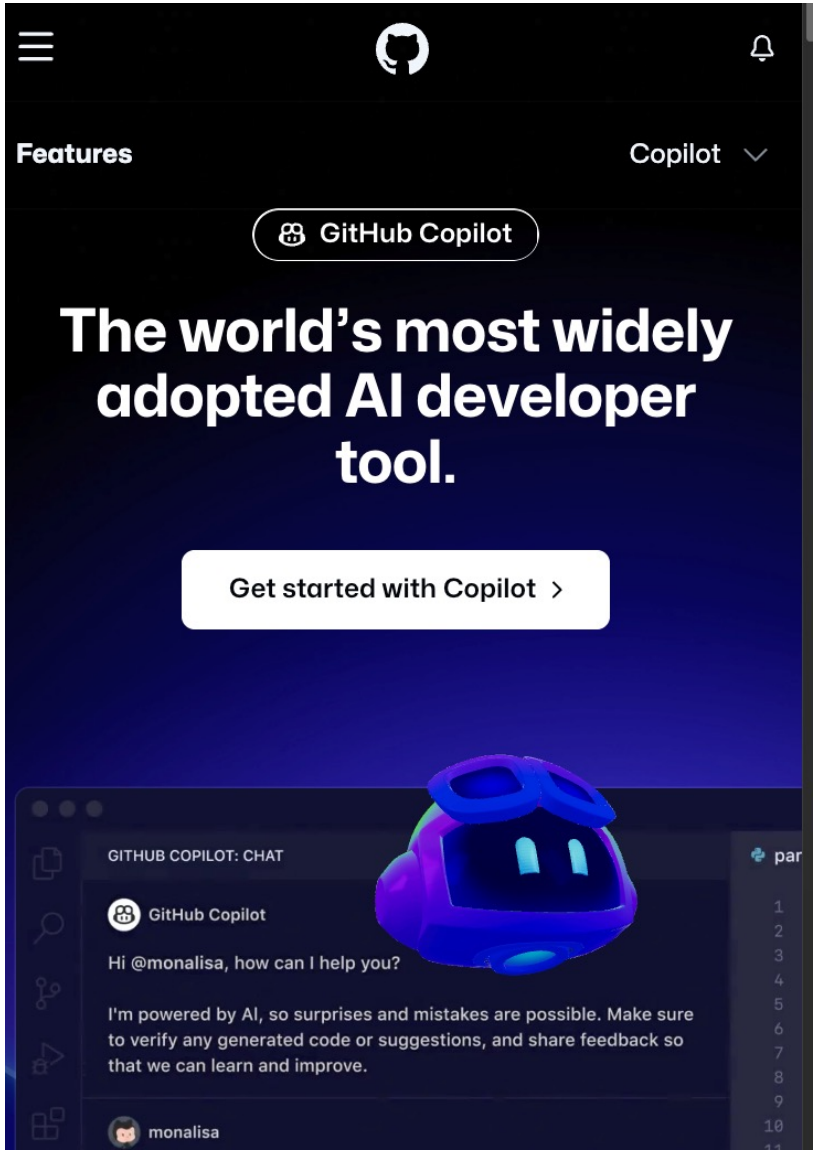
- You are allowed to use ChatGPT and other AI tools to facilitate your learning.
- Learning detailed algorithum is beyond this course. At this point, you have to know only one thing:
  - **They are very useful; sometimes they are correct, but sometimes NOT. They are statistical or machine learning "models," so their answers are basically PREDICTIONS based on existing knowledge.**

ChatGPT

https://chat.openai.com/

※ Copilot with VS code, RStudio or other editors.

- Copilot (with VS code and RStudio)
  - AI programming assistant
  - GitHub Copilot is powered by a generative AI model developed by **GitHub**, **OpenAI**, and **Microsoft**.
  - Can be used with VS code, RStudio, or other editors.
  - Programming-oriented chatGPT?
- **You may use it** and **please declare** that you used copilot in your assignments.



https://github.com/features/copilot

Information for assignments

In-Class Assignments

- In-Class Assignment (40%)
  - **At most, there are 8 assignments. Each assignment will be 5 points** (if < 8 assignments, points allocated to each assignment will be adjusted). In each assignment, you will have **3-4 small tasks**.
- Final Project (60%)

- Schedule and rules for the assignments:
  - I will upload a PDF that explains assignments **every Monday or Tuesday** (after the first lecture in each week)
  - I will explain dataset and you can work on it **on Wednesday (in a lecture and tutorial)**. You may work on them before Wednesday if you are able to do so.
  - **The deadline is at 23:59 each Thursday**.
  - You may use Google search, ChatGPT, etc… But in the lecture, I will encourage you to do the tasks by yourself in the first 30 min, and then you may discuss with your colleagues.

In-Class Assignments

How to submit:

- **via Canvas** (but note that this course is a new one, there might be some adjustment…)
- **ONE Jupyter notebook file (.jpynb file)** or **python code (.py)**

Marking criteria:

- I will run your code in my local environment
- **Figures are produced and no flaw** [**5 pt.**]
- **Figures are generally OK, but need some improvement**, e.g., color and layout [**4 pt**]
- **Submitted your code but fail to produce figures** [**1-3 pt.**]
- **Not submitted by the deadline** [**0 pt.**]
- **Please declare in your code if you use ChatGPT or other AI tools**. Also please briefly explain how you utilize the AI tools (e.g., 1-2 sentences as comments in your python code).
- **DO NOT copy and paste other students' code or file**. If I found completely identical code, **all the students who submitted identical code will get 0 pt**.
- If you submit optional task(s), you may earn 1 more point (but the total point can not be more than 5 pt.)

The Basics of Python

Let's review the basic of Python!

https://github.com/ong8181/DASC3240

Scripts for hands-on in DASC3240

This repository includes scripts for DASC3240 offered in Spring semester 2024 at HKUST.

License

See LICENSE.

Open Jupyter notebook in your local PC

If Jupyter Notebook is already installed in your PC, the most straightforward way to run the scripts is probably downloading all the files in this repository and open them using jupyter notebook. Alternatively, you can open them in Google Colab, but reading demo data is a bit tricky.

Open Jupyter Notebook in Google Colab

Materials for lectures

- Open in Colab L04_ReviewPython.ipynb

- If you already have jupyter notebook and Python, please download the codes.

- If you do not have, please click "Open in Colab" button.